

Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion

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Abstract

Scene graphs represent visual scenes as objects and their relationships. While considerable research has explored predicting scene graph sequences from images or videos, generating such sequences from text remains underexplored. In this work, we introduce a discrete diffusion framework for text-conditioned scene graph sequence generation, focusing on visual scenes centered on human-object interactions. The proposed framework represents each sequence as a fixed-slot discrete state, where each slot encodes the state of an object over time, and employs a Transformer-based denoiser to generate scene graph sequences aligned with the input text. The denoiser jointly leverages within-frame slot attention, cross-frame temporal attention, and text cross-attention to capture inter-object relationships, temporal dynamics, and text-graph correspondence. We compare the proposed framework with diverse baselines, including retrieval-based methods, scene graph parsers, large language models, sequence-to-sequence models, and non-diffusion generative models. Experimental results show that our framework improves both text-graph alignment and the accuracy of temporal dynamics represented in scene graph sequences, including object appearance, persistence, and disappearance.

CCS Concepts

• **Computing methodologies** → **Scene understanding; Structured outputs; Natural language processing; Neural networks.**

Keywords

Scene graph sequence generation, Discrete diffusion, Text-conditioned generation, Text-to-graph generation

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1 Introduction

Scene graphs represent objects and their relations as nodes and edges, providing a structured semantic representation for visual understanding, retrieval, and relational reasoning [16, 17, 35]. Recent work has extended static graphs to scene graph sequences, which

capture object states and interactions evolving over an event [7, 15, 21]. In this work, we study text-conditioned scene graph sequence generation. Specifically, we focus on an anchor-centric setting, where each scene contains a central object, referred to as an anchor, and the goal is to generate a scene graph sequence that describes the relations between the anchor and its surrounding objects over time. For example, in the Action Genome dataset [15], each scene graph describes the relations between a human anchor and the surrounding objects in the scene. This anchor-centric view naturally arises in many scene graph applications, where understanding an event often requires modeling how a primary entity interacts with its context. Existing studies on scene graph sequences mainly predict graphs from video frames or detected objects, leaving generation from natural language alone underexplored. Unlike graph prediction from video, which requires visual observations, text-conditioned generation can produce a structured temporal representation directly from descriptions, including hypothetical or user-edited events for which no corresponding video exists. Converting these descriptions into graph sequences provides an event-level representation that can support video retrieval, controllable video generation, and temporal reasoning benchmarks [9].

Static text-to-scene-graph methods produce only a single graph, and therefore cannot capture how objects and their relations change over time [19, 20, 31]. In contrast, our task is to generate a time-ordered graph sequence from language, where each graph consists of categorical object and relation labels. This requires a generative model over discrete structured states. Discrete diffusion has recently shown promise for generating categorical structures such as graphs [2, 33], but language-conditioned scene graph sequence generation remains underexplored.

We first formulate this problem as a learning-based generative task. We train a discrete diffusion framework on text-graph sequence pairs, where each pair links a sentence to the scene graph sequence from the video segment described by that sentence. Our framework represents each sequence as a fixed set of object slots, where each slot tracks one object's state over time. A Transformer denoiser learns to recover the clean graph sequence from a noisy version by combining within-frame, cross-frame, and text attention. We further strengthen text-conditioned generation with classifier-free guidance and text-aware loss boosting [14].

Our contributions are threefold: (a) we formulate text-conditioned scene graph sequence generation as probabilistic discrete structure generation for temporally evolving interactions; (b) we introduce an anchor-centric fixed-slot representation and a diffusion Transformer denoiser for frame-to-frame object and relation states; and (c) we evaluate against retrieval, parser, LLM, seq2seq, and non-diffusion generative baselines, showing improved text-graph alignment and better capture of object states and changes over time.



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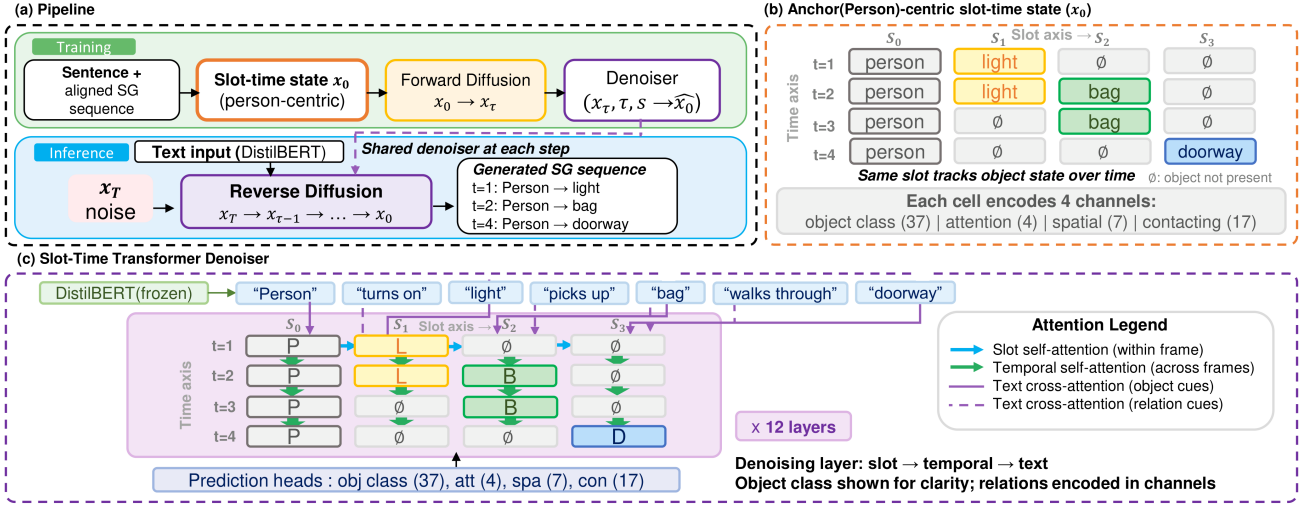


Figure 1: Overview of the proposed framework: (a) discrete diffusion training/inference pipeline, (b) person-centric slot-time state, and (c) Slot-Time Transformer Denoiser. The denoiser applies slot, temporal, and text attention over the slot-time state.

2 Related Work

2.1 Scene Graph Generation

Scene graph sequence generation aims to represent changes in object interactions over time and has been mainly studied on Action Genome [15]. Prior video scene graph methods improve temporal relation prediction but still rely on visual inputs such as frames or detected objects [7, 10, 18, 21]. Textual scene graph parsers such as FACTUAL and DiscoSG construct static graphs from text [19, 20]. DySen-VDM uses LLM-generated dynamic graphs as conditions for video diffusion rather than as the target output [9]. In contrast, we directly generate scene graph sequences from natural language.

2.2 Graph Diffusion Models

Diffusion models are effective for continuous generation, especially images [13, 30], and have been extended to discrete state spaces. D3PM defines diffusion over discrete variables [2], and DiGress applies discrete diffusion to categorical graph generation [33]. Other paradigms for discrete structure generation include autoregressive graph models [34], variational graph models [29], adversarial graph models [8], and masked refinement models [4, 12], which we compare against as baselines. Scene graph diffusion work has mainly used graphs as conditions for image generation or addressed video-input-based graph prediction [5, 28]; our goal is text-conditioned graph sequence generation.

3 Method

3.1 Problem Formulation

Figure 1 provides an overview of the proposed framework. Given a natural language sentence s , our goal is to generate a scene graph sequence $\mathcal{G} = \{G_1, G_2, \dots, G_L\}$. We formulate the task in an anchor-centric interaction graph format, where each frame describes the relations between a central anchor and its surrounding objects over time. In Action Genome, the anchor is the person; therefore, we

model person-object interactions rather than all pairwise object-object relations. For each object slot, we predict the object class and its relation to the anchor.

Anchor-centric slot representation. Each sample is represented as an $L \times N$ slot-time structure, where L is the maximum sequence length, N is the number of slots, slot 0 is the anchor, and the remaining $N-1$ slots are object slots. Each slot-time position contains four channels: object class x^{node} , attention relation x^{att} , spatial relation x^{spa} , and contacting relation x^{con} . The object, attention, and spatial channels are categorical, while contacting is multi-label binary because an object can have multiple contacting relations at the same time. The clean state is $\mathbf{x}_0 = (x_0^{\text{node}}, x_0^{\text{att}}, x_0^{\text{spa}}, x_0^{\text{con}})$. Slots are assigned once per sample using a deterministic first-appearance rule computed over the full annotated interval before temporal subsampling. Each object class is assigned to the next available non-anchor slot in the order of its first appearance, with the assignment fixed across all frames. As a result, each slot consistently tracks the corresponding object’s state over time, allowing appearance, persistence, disappearance, and relation changes to be modeled as slot-level transitions without explicit tracking IDs.

3.2 Discrete Diffusion for Graph Sequences

We use discrete diffusion to generate the sentence-conditioned slot-time state. The model corrupts the clean state $\mathbf{x}_0$ into $\mathbf{x}_\tau$ and learns to recover $\mathbf{x}_0$ from $(\mathbf{x}_\tau, \tau, s)$. For each categorical channel $c \in \{\text{node}, \text{att}, \text{spa}\}$, we use marginal transition-based discrete diffusion [2, 33]. Let m_c be the empirical label distribution of channel c from the training set. Each categorical value is represented as a one-hot row vector. The channel-specific one-step transition matrix Q_τ^c and cumulative transition matrix $\bar{Q}_\tau^c$ are

$$Q_\tau^c = (1 - \beta_\tau)I + \beta_\tau \mathbf{1} m_c^\top, \quad \bar{Q}_\tau^c = \bar{\alpha}_\tau I + (1 - \bar{\alpha}_\tau) \mathbf{1} m_c^\top, \quad (1)$$

where β_τ is the noise rate, $\bar{\alpha}_\tau$ is the cumulative signal rate, I is the identity matrix, and $\mathbf{1}$ is the all-ones vector. The forward process is

$$q(\mathbf{x}_\tau^c | \mathbf{x}_0^c) = \text{Cat}(\mathbf{x}_\tau^c; \mathbf{x}_0^c \bar{Q}_\tau^c). \quad (2)$$

For the multi-label contacting channel, we use independent binary diffusion: each label is preserved with probability $\bar{\alpha}_\tau$ and otherwise replaced by uniform binary noise, i.e., Bernoulli(0.5).

The denoiser predicts the clean state:

$$\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_\tau, \tau, s). \quad (3)$$

At each reverse step, we form the standard D3PM reverse posterior from $\hat{\mathbf{x}}_0$ and the transition matrices, and sample $\mathbf{x}_{\tau-1}$.

The training objective is $\mathcal{L} = \mathcal{L}_{\text{node}} + \mathcal{L}_{\text{att}} + \mathcal{L}_{\text{spa}} + \mathcal{L}_{\text{con}}$, using cross-entropy for categorical channels and binary cross-entropy for contacting. Relation losses are computed only where the anchor and object are non-null. We use inverse-frequency class weighting for label imbalance and TextBoost to increase the loss weights for slots whose object classes are mentioned in the input sentence. Object mentions are identified using a hand-written alias dictionary covering all 36 Action Genome classes with whole-word matching on lowercased text.

3.3 Slot-Time Transformer Denoiser

We design a Transformer denoiser that applies self-attention along the slot and time axes, which we call the Slot-Time Transformer Denoiser. Given a noisy slot-time state $\mathbf{x}_\tau$ and the input sentence, it predicts the clean state $\hat{\mathbf{x}}_0$ on the anchor-centric fixed-slot grid, where each token represents one object slot at one frame. The model uses standard Transformer attention [32]. Its slot-time factorization is inspired by video Transformers that separate attention within a frame from attention across time [1, 3]. Each token combines embeddings of the four state channels—object class, attention, spatial, and contacting—with slot, frame, and diffusion-step embeddings. A frozen DistilBERT encoder [27] encodes the input sentence, and its hidden states are used for text cross-attention. Slot attention performs self-attention over the N slots independently within each frame, while temporal attention performs self-attention over the L timesteps independently for each fixed slot. Text cross-attention then allows each slot-time token to attend to these token-level text representations. These three attention modules are applied sequentially in each denoising layer, in the order of slot attention, temporal attention, and text cross-attention. They capture object context within a frame, state transitions over time, and text-graph alignment, respectively. Prediction heads output categorical logits for the object class, attention, and spatial channels, and per-label binary logits for the multi-label contacting channel.

3.4 Classifier-Free Guidance and Sampling

During training, we drop the text condition with a fixed probability by replacing the sentence with an empty string. This lets the denoiser learn both conditional predictions, using the input sentence, and unconditional predictions, without the sentence. At inference time, we apply classifier-free guidance [14] to the predicted logits:

$$\tilde{f}_\theta = f_\theta(\mathbf{x}_\tau, \tau, \emptyset) + \gamma (f_\theta(\mathbf{x}_\tau, \tau, s) - f_\theta(\mathbf{x}_\tau, \tau, \emptyset)), \quad (4)$$

where γ is the guidance scale. For sampling, we initialize $\mathbf{x}_T$ from the marginal distribution and run K reverse steps using the $\hat{\mathbf{x}}_0$ -parameterized posterior. The case $K = 1$ corresponds to direct prediction from maximum noise. Main results use development-selected $K = 2$ with a 3-sample ensemble: we average final-step $\hat{\mathbf{x}}_0$ logits from independent samples and decode by argmax for categorical channels and a 0.5 threshold for contacting. No reference sequences or test labels are used.

4 Experiments

4.1 Experimental Setup

Dataset. We construct text-graph sequence pairs from two annotations on Charades videos. Charades-STA [11] provides sentences with temporal intervals $[t_s, t_e]$ describing video segments, and Action Genome [15] provides frame-level person-centric scene graphs. For each sentence, we collect Action Genome frame graphs whose timestamps $t = \text{frame_idx}/\text{fps}$ fall within $[t_s, t_e]$; these graphs form the sentence-aligned reference sequence. Thus, each example pairs a text description with the person-centric graph sequence observed in that segment. We discard sentences with no annotated frames and subsample or pad each sequence to $L = 16$. The official video-ID-disjoint Charades-STA split yields 9,986 training and 3,071 test pairs, with 500 training pairs held out for development. Action Genome provides 36 object classes and 26 relations: 3 attention (Att), 6 spatial (Spa), and 17 contacting (Con). We add null states for empty object slots and categorical relation channels, and use $N = 14$ slots.

Metrics. Each output is compared with the sentence-aligned Action Genome sequence constructed above, used as the reference for evaluation. This reference is one observed version of the sentence in the video, not the only possible correct graph sequence. For fair comparison, all methods use the same fixed $L = 16$ frame grid and are converted into frame-level tuples: object tuples (t, c) indicate object class c at frame t , and relation tuples (t, c, k, v) indicate that object class c has relation value v of type k with the anchor at frame t . All reported scores except Diversity are F1 scores.

We report three groups of metrics, ordered by their role in our task. First, **text-grounded** metrics, TG-Obj and TG-Rel, ask whether the generated sequence matches the input sentence. They compute F1 over object and relation tuples whose labels are mentioned in the sentence, identified by a fixed word-to-label mapping shared by all methods. Since tuples include t , matching requires the described objects and interactions to appear at the correct timesteps. Second, **temporal-state** metrics test whether a model keeps objects stable and captures state changes over time. SK measures whether a slot that stays non-empty and unchanged between adjacent reference frames is also kept in the prediction; Class-SK additionally requires the predicted object class to match. Appear and Vanish report one-vs-rest F1 for newly appearing and disappearing objects, and Change-only pools appearance, disappearance, and object-change transitions. Third, **full-reference** metrics, Obj and Rel, compute F1 over all object and relation tuples in the aligned reference, including elements not explicitly named in the sentence. These metrics are not our primary target; rather, they measure how much of the aligned Action Genome graph sequence is recovered and whether the model learns plausible graph patterns beyond

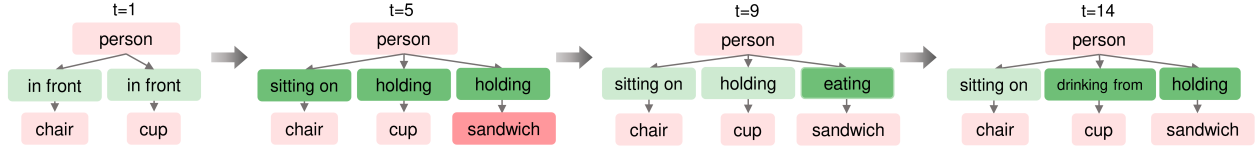


Figure 2: Example output of Ours-144M for the input sentence: “person sits on the chair eats a sandwich and drinks from a cup.”

explicitly mentioned labels. Att, Spa, and Con decompose Rel by relation type, and Diversity is the average pairwise object Jaccard distance over five generations for the same input.

Implementation. The denoiser uses frozen DistilBERT [27]; its parameters are excluded from model size. The 144M variant has hidden size 768, 12 layers, and 12 heads. We train with a 200-step cosine discrete diffusion objective [2, 33], followed by relation-weighted fine-tuning with relation loss weight 2.0. On the development set, we choose inference hyperparameters that give high Rel F1 while keeping SK high; the final setting uses $K = 2$, guidance scale $\gamma = 1.2$, a 3-sample ensemble, and no slot deduplication.

Baselines. Baselines include TextMatch, BM25 [26], and SBERT [25] for retrieval; FACTUAL [19] for static textual scene graph parsing; Flan-T5-large [6], GPT-4o-mini [22], GPT-4o [23], and DySen-VDM-style prompting [9] for zero-shot prompting; fine-tuned T5-small/base [24]; and our 24M/144M models. Non-temporal outputs are deterministically converted to frame-level tuples, with static graphs replicated over frames and non-slot outputs assigned to slots by class name for SK. For Table 2, we implement task-adapted 24M generators under the same fixed-slot representation, frozen text encoder, split, and vocabulary.

4.2 Results and Analysis

Main results. Figure 2 visualizes representative timesteps where the generated graph changes, and Table 1 reports full-test results. Object-only matching and zero-shot LLM baselines recover text-mentioned objects but poorly localize temporal relations, yielding near-zero TG-Rel; DySen-style prompting obtains non-trivial TG-Rel but low Rel and SK. Ours-24M improves TG-Obj, TG-Rel, SK, Obj, and Rel over T5-base with about nine times fewer trainable parameters, and its TG-Rel gain is statistically significant (0.516 vs. 0.464; $\Delta = +0.052$, paired video-cluster bootstrap 95% CI [+0.024, +0.078], permutation $p < 0.001$). Ours-144M gives the best overall Rel, although spatial relations remain challenging.

Comparison with alternative generative models. Table 2 compares 24M generators under the same fixed-slot setting. AR-GraphSeq and GraphSeq-GAN show weak accuracy, while VAE and Mask-Graph are stronger non-diffusion alternatives. Ours-24M achieves the best TG-Obj, TG-Rel, SK, Obj, and Rel, showing that marginal-transition diffusion improves text-grounded generation, temporal consistency, and reference-based graph recovery. Although AR-GraphSeq has the highest raw diversity, its accuracy is much lower; among accurate generators, Ours provides the best accuracy–diversity trade-off.

State-transition analysis. Table 3 checks whether the high SK comes from fixed-slot formatting. For this analysis, T5-base-slot is T5-base serialized into the same fixed-slot format.

Table 1: Main results on the full test set ($n = 3,071$).

Category	Model	Params	TG-Obj	TG-Rel	SK	Obj	Rel	Att	Spa	Con
Training-free	TextMatch	–	0.781	0.000	0.495	0.433	0.211	0.187	0.306	0.143
	BM25	–	0.780	0.370	0.553	0.372	0.221	0.197	0.248	0.219
	SBERT	–	0.826	0.400	0.550	0.405	0.240	0.213	0.273	0.235
SG parser	FACTUAL	250M	0.814	0.389	0.405	0.387	0.037	0.008	0.001	0.091
LLM	Flan-T5-L	783M	0.826	0.000	0.249	0.144	0.070	0.082	0.086	0.042
	GPT-4o-mini	–	0.874	0.018	0.594	0.417	0.201	0.201	0.281	0.123
	GPT-4o	–	0.862	0.018	0.607	0.447	0.210	0.210	0.294	0.129
	DySen-style	–	0.700	0.398	0.387	0.339	0.068	0.025	0.037	0.129
Seq2seq	T5-small	60M	0.366	0.224	0.431	0.239	0.151	0.133	0.174	0.145
	T5-base	222M	0.787	0.464	0.432	0.442	0.285	0.247	0.331	0.278
Ours	Ours-24M	24M	0.882	0.516	0.982	0.523	0.305	0.288	0.286	0.335
	Ours-144M	144M	0.886	0.508	0.978	0.519	0.309	0.292	0.298	0.332

Table 2: Comparison with 24M alternative generators.

Paradigm	Model	TG-Obj	TG-Rel	SK	Obj	Rel	Diversity
Autoregressive	AR-GraphSeq	0.699	0.357	0.512	0.365	0.182	0.524
Adversarial	GraphSeq-GAN	0.801	0.295	0.706	0.431	0.194	0.020
Variational	GraphSeq-VAE	0.875	0.480	0.981	0.506	0.289	0.060
Masked refinement	Mask-Graph	0.872	0.460	0.969	0.508	0.278	0.000
Diffusion	Ours-24M	0.882	0.516	0.982	0.523	0.305	0.359

Table 3: Checks against slot and transition artifacts.

Model	SK	Class-SK	Appear	Vanish	Change-only
T5-base	0.432	0.304	0.000	0.000	0.000
T5-base-slot	0.305	0.211	0.000	0.000	0.000
Ours-144M	0.978	0.698	0.996	0.996	0.888

Ours is much stronger on Class-SK and on the explicit transition metrics, Appear, Vanish, and Change-only, whereas both T5 variants score zero on all three. This means that T5 fails to represent when objects appear, disappear, or change, and the T5-base-slot row shows that simply using the fixed-slot output format does not fix the issue. Additional slot-assignment checks lead to the same conclusion: oracle slot assignment keeps T5-base near 0.43 SK, and a slot-free graph-edit check, which compares frame-to-frame object changes without using slot identities, gives T5-base 0.000/0.000 appear/vanish F1 versus 0.287/0.288 for Ours-144M.

Relation analysis. Charades-STA provides limited relation cues: only 44.6% of test sentences mention relation words, and only 6.6% mention spatial words. Despite this, Ours-144M gives a modest but reliable Rel improvement over T5-base (0.309 vs. 0.285; +0.024, 95% CI [+0.016, +0.032], permutation $p < 0.001$). The gain appears both when relations are explicitly mentioned and when they are not (0.335/0.286 vs. 0.307/0.265), suggesting that the model is not only copying relation words. By type, the improvement mainly comes from attention (+0.045) and contacting (+0.054) relations; spatial remains lower (−0.033), likely because spatial cues are rarely stated in the text.

4.3 Ablation Study

Table 4: Ablation study on the full test set ($n = 3,071$).

Config.	TG-Obj	TG-Rel	SK	Obj	Rel
24M backbone	0.783	0.409	0.952	0.411	0.245
+ Classifier-free guidance	0.850	0.429	0.952	0.470	0.233
+ Scale to 144M	0.860	0.412	0.907	0.485	0.211
+ TextBoost	0.863	0.432	0.985	0.479	0.260
+ Rel-weighted finetune	0.869	0.409	0.981	0.499	0.278
+ Dev-selected $K=2$ (1-sample)	0.885	0.489	0.973	0.506	0.292
+ 3-sample ensemble	0.886	0.508	0.978	0.519	0.309
w/o temporal attention	0.882	0.497	0.872	0.513	0.304

Table 4 adds components step by step, with the final row removing temporal attention. Classifier-free guidance improves text-grounded F1, whereas scaling alone does not improve relation F1. TextBoost improves TG-Rel, relation-weighted fine-tuning improves Rel, and the development-selected $K = 2$ sampler gives the largest relation-metric gain; the 3-sample ensemble further improves TG-Rel, Obj, and Rel. Removing temporal attention mainly hurts temporal consistency, with SK dropping from 0.978 to 0.872 while other metrics change only slightly. Together with Table 2, these results illustrate two practical benefits of diffusion: classifier-free guidance helps the output better match the input sentence, and stochastic sampling produces diverse plausible graph sequences for the same text.

5 Conclusion

We formulated text-conditioned scene graph sequence generation as probabilistic discrete structure generation with an anchor-centric fixed-slot representation. Experiments show that the proposed diffusion framework improves text-graph alignment and better captures object states and changes over time than diverse baselines. By studying the task in a fixed-vocabulary human-object interaction setting, this work provides a foundation for future research extending text-to-graph-sequence generation to richer interaction structures, open-vocabulary graphs, broader evaluation settings, and applications in video retrieval, generation, and temporal reasoning.

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GenAI Usage Disclosure

The authors used generative AI tools to edit and enhance text, similar to grammar- and clarity-improvement tools. The tools were used for language polishing, sentence-level editing, and improving clarity. All technical content, experimental results, analysis, and final claims were developed, checked, and verified by the authors, who take full responsibility for the paper.

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